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CANDIDATE NAME

CLASS

ADMISSION NUMBER

2020 Preliminary Examination
Pre-University 3

MATHEMATICS **9758/01**

Paper 1 **14 Sep 2020**

3 hours

Candidates answer on the Question Paper.

Additional Materials: List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your admission number, name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.
Give your answers in the spaces provided in the Question Paper.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 100.

Qn No.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	*	Total
Score													
Max Score	**5**	**5**	**8**	**8**	**9**	**9**	**10**	**11**	**11**	**12**	**12**		**100**

This document consists of **6** printed pages.

1 Sketch the graph of $y = \ln\left(x^2 - 4\right)$, stating the equations of any asymptotes. [2]

By adding a suitable curve on your graph above, solve the inequality

$$\ln\left(x^2 - 4\right) + x^2 - 8 \le 0 .$$ [3]

2 A spherical balloon has radius r cm. The surface area is increasing at a rate of π cm^{2}s^{-1}.
[It is given that a sphere of radius r has surface area $4\pi r^2$ and volume $\frac{4}{3}\pi r^3$.]

(i) Find an expression for $\frac{\mathrm{d}r}{\mathrm{d}t}$ in terms of r. [2]

(ii) Hence find the rate of change of the volume of the balloon at the instant when the volume of the balloon is $\frac{288}{\pi^2}$ cm^3. [3]

3 A sequence $u_1,\ u_2,\ u_3,\ \ldots$ is given by

$$u_r = \frac{1}{(r+1)!} \quad \text{for } r \ge 1.$$

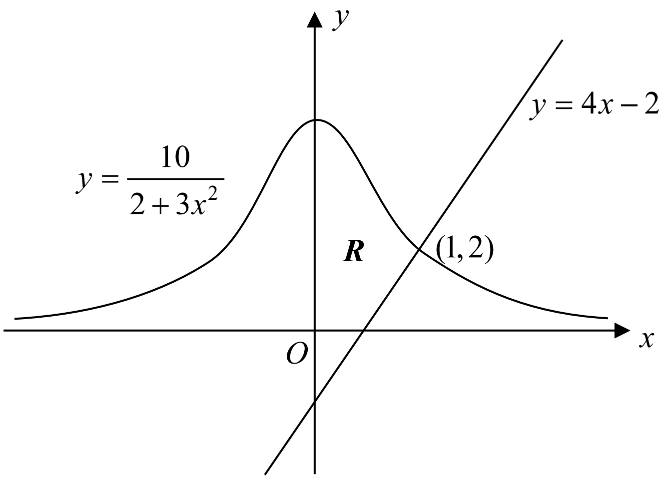
(i) Show that $u_r - u_{r+1} = \frac{r+1}{(r+2)!}$. [1]

It is given that $S_n = \sum_{r=1}^{n} \frac{r+1}{(r+2)!}$.

(ii) Hence find an expression for S_n in terms of n. (There is no need to express your answer as a single algebraic fraction.) [4]

(iii) Find the smallest value of n for which S_n is within 10^{-4} of the sum to infinity. [3]

4 The diagram below shows the region R which is bounded by the curve $y = \dfrac{10}{2+3x^2}$, the line $y = 4x - 2$ and the y-axis.



(i) Find the area of the region R, giving your answer correct to 3 decimal places. [3]

(ii) Find the exact volume of the solid of revolution formed when R is rotated through 2π radians about the y-axis. [5]

5 The curve C_1 has equation $y = x^2$ and the curve C_2 has equation $y = (ax-b)^2 + c$, where a, b and c are positive constants.

(i) Describe a sequence of transformations that transforms C_1 onto C_2. [3]

(ii) The point $(1,\ 1)$ on C_1 corresponds to the point (p, q) on C_2 after the sequence of transformations in part **(i)**. It is also given that C_2 passes through the point $(0, r)$. Find p, q and r in terms of a, b and c. [3]

(iii) Given that C_2 passes through the point $(2, 12)$ and has a turning point at $\left(\dfrac{1}{2}, 3\right)$. Find the values of a, b and c. [3]

6 It is given that $y=\sqrt{e^x \cos x}$.

(i) Show that $2y\dfrac{dy}{dx}=y^2-e^x \sin x$. [2]

(ii) By further differentiation of the result in part **(i)**, find the Maclaurin series for y, up to and including the term in x^2. [4]

(iii) Using standard series from the List of Formulae (MF26), expand $\sqrt{e^x \cos x}$ as far as the term in x^2 and verify that the same result is obtained in part **(ii)**. [3]

7 With respect to the origin O, the position vectors of the points A and B are $5\mathbf{i}+2\mathbf{j}+\mathbf{k}$ and $\mathbf{i}-2\mathbf{j}+4\mathbf{k}$ respectively. The plane Π_1 has equation $2x+y-4z=8$.

(i) Verify that A lies on Π_1. [1]

(ii) Find the coordinates of the foot of perpendicular from B to Π_1. [3]

(iii) Hence, or otherwise, find the perpendicular distance from B to Π_1. [2]

The planes Π_2 and Π_3 have equations $\mathbf{r}\bullet\begin{pmatrix}1\\-3\\1\end{pmatrix}=2$ and $\mathbf{r}\bullet\begin{pmatrix}-1\\1\\3\end{pmatrix}=5$.

(iv) Find the acute angle between Π_2 and the normal of Π_3. [2]

(v) It is given that the three planes Π_1, Π_2 and Π_3 intersect at a common point of intersection. By considering the Cartesian equations of Π_1, Π_2 and Π_3 and forming a system of linear equations, find the position vector of the point of intersection of the three planes. [2]

8 The function f is defined by

$$\mathrm{f}: x \mapsto \frac{1}{4-x^2},\ x\in\mathbb{R},\ x\geq 0,\ x\neq 2.$$

(i) Sketch the graph of $y=\mathrm{f}(x)$, stating clearly the coordinates of the y-intercept and the equations of the asymptotes. Hence explain why f^{-1} exists. [3]

(ii) Find $\mathrm{f}^{-1}(x)$ and write down the domain of f^{-1}. [3]

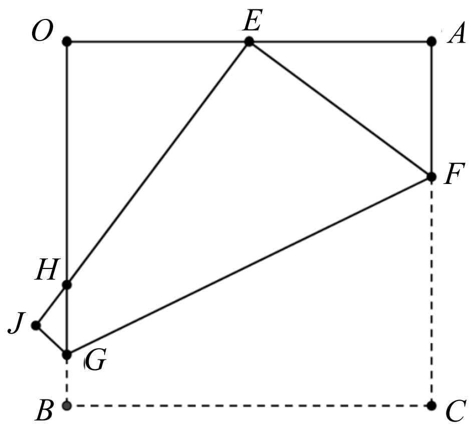
(iii) Sketch on the same diagram in part **(i)**, the graph of $y = \mathrm{f}^{-1}(x)$, giving the equations of any asymptotes.

Explain why the x-coordinates of the points of intersection between the graphs of $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$ satisfy the equation

$$x^3 - 4x + 1 = 0.$$ [4]

Hence find the x-coordinates of the points of intersection between the graphs of $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$. [1]

9 In the figure below, $OACB$ is a square with sides of length 1 unit each and E is the midpoint of OA. The figure shows the configuration when the square $OACB$ is folded such that C is now placed at point E and B is now placed at point J.



The position vectors of A and B are $\mathbf{a}$ and $\mathbf{b}$ respectively with respect to the origin O.

(i) Let the length of AF be x. By using Pythagoras' Theorem, find the value of x. Hence show that $\overrightarrow{EF} = \frac{1}{2}\mathbf{a} + \frac{3}{8}\mathbf{b}$. [3]

(ii) It is given that $\overrightarrow{EH} = -\frac{1}{2}\mathbf{a} + \mu\mathbf{b}$. By considering the scalar product of $\overrightarrow{EF}$ and $\overrightarrow{EH}$, find the value of μ. [4]

P is the point of intersection of CB produced and EH produced.

(iii) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, vector equations of lines BC and EH. [2]

(iv) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, the position vector of P. [2]

10 An ecologist and an entomologist are investigating the change in the population of a particular species of insects of size x thousand at time t months.

The ecologist suggests that x and t are related by the differential equation $\dfrac{\mathrm{d}^2x}{\mathrm{d}t^2}=9-5t$.

(i) Find the general solution of this differential equation. [2]

(ii) Given that there are 1000 insects initially and that there is no change in population on the third month, find the particular solution of this differential equation. [3]

The entomologist suggests that the rate of change of the insect population is inversely proportional to the square of the insect population.

(iii) Write down a differential equation relating x and t. [1]

(iv) Given that there are 1000 insects initially and that the insect population on the first month is 4000, find t in terms of x.
Hence find the time taken for the population to reach 16000. [6]

11 Arthur uses a special type of clay to make a multi-layered structure in which each layer is in the shape of a cylinder. The base layer has a radius of 60 cm and a height of 20 cm. The radius of each layer above is 3 cm less than the radius of the previous layer. The height of each layer remains the same.

(i) Show that the maximum number of layers that a structure can have is 20. [2]

(ii) Ribbon is pasted around the circumference of each layer of a structure. Find the total length of ribbon required for a 20-layered structure. [3]

(iii) Show that the total volume of a 20-layered structure is given by

$$p\pi\sum_{n=1}^{20}\left[\mathrm{f}(n)\right]^2,$$

where p is a constant to be determined and the function $\mathrm{f}(n)$ is to be found. [3]

Given that Arthur has 1 500 000 cm^3 of clay, explain if Arthur is able to construct a 20-layered structure. [1]

To make a multi-layered structure more appealing, the height of each layer is now varied such that the height of the layer above is 80% of the height of the previous layer.

(iv) Assuming that the height of the base layer is still 20 cm, determine the maximum number of layers a structure can have if the height of the structure is to be at most 85 cm. [3]

End of Paper